Prompt 1: To create a MCP server, Tools along with DB

You are a Senior Solution Architect.

Create a complete professional architecture diagram for the following project.

PROJECT NAME: Company Data Assistant using MCP (Model Context Protocol)

PROJECT OVERVIEW:

The system allows users to ask natural language questions about company data.

Claude Desktop acts as the AI client.

Claude communicates with a custom MCP Server developed in C# (.NET 8).

The MCP Server exposes tools that access a SQLite database.

Claude automatically selects the correct tool based on user questions.

The MCP Server retrieves data from SQLite and returns results.

Claude formats the returned data into human-readable responses.

TECHNOLOGY STACK

Frontend Client:
- Claude Desktop

Backend:
- MCP Server
- C# (.NET 8)

Database:
- SQLite

Development Tools:
- VS Code
- Git
- Codex

TEAM STRUCTURE

Team Member 1:
Database Engineer

Responsibilities:
- Database Design
- Table Creation
- Sample Data Population
- Query Validation

Team Member 2:
MCP Developer

Responsibilities:
- MCP Server Development
- Tool Registration
- Tool Descriptions
- MCP Configuration

Team Member 3:
Backend Tool Developer

Responsibilities:
- Database Service Layer
- SQL Execution Layer
- Error Handling
- Logging

Team Member 4:
Integration & Testing Engineer

Responsibilities:
- Claude Desktop Integration
- MCP Configuration Testing
- End-to-End Testing
- Demo Preparation

DATABASE SCHEMA

Departments

DepartmentId (PK)
DepartmentName

Employees

EmployeeId (PK)
Name
Email
DepartmentId (FK)

Projects

ProjectId (PK)
ProjectName
Status

EmployeeProjects

EmployeeId (FK)
ProjectId (FK)

MCP TOOLS

1. list_tables()
Purpose:
Return all database tables

2. get_schema(tableName)
Purpose:
Return schema of specified table

3. run_select(query)
Purpose:
Execute SELECT queries only

4. get_employee_count()
Purpose:
Return employee count

5. get_employees_by_department(departmentName)
Purpose:
Return employees in a department

6. get_department_summary()
Purpose:
Return department statistics

7. get_active_projects()

Purpose:
Return active projects

8. get_employee_projects(employeeName)
Purpose:
Return projects assigned to employee

DATA FLOW

Step 1:
User asks question in Claude Desktop

Example:
"How many employees are in IT?"

Step 2:
Claude analyzes the question

Step 3:
Claude selects appropriate MCP Tool

Example:
get_employees_by_department()

Step 4:
Claude sends tool request to MCP Server

Step 5:
MCP Server invokes corresponding Tool

Step 6:
Tool calls Database Service

Step 7:
Database Service executes SQL against SQLite

Step 8:
SQLite returns result

Step 9:
Result returned to MCP Tool

Step 10:
MCP Tool returns structured data to Claude

Step 11:
Claude formats response

Step 12:
Claude displays human-readable answer to user

ARCHITECTURE LAYERS

Layer 1:
User Layer

Layer 2:
AI Layer
- Claude Desktop

Layer 3:
MCP Layer
- MCP Server
- Tool Registry
- Tool Dispatcher

Layer 4:
Business Logic Layer
- Database Service
- Query Validation
- Logging
- Error Handling

Layer 5:
Data Layer
- SQLite Database

SECURITY COMPONENTS

- Read Only Access
- SELECT Queries Only
- Block INSERT
- Block UPDATE
- Block DELETE
- Block DROP
- Block ALTER

- Input Validation
- Error Logging

GENERATE

1. High-Level Architecture Diagram
2. Detailed Component Diagram
3. Data Flow Diagram
4. Deployment Diagram
5. Team Responsibility Diagram
6. Sequence Diagram showing:
   User → Claude → MCP → Tool → Database → Claude → User
7. Professional Architecture suitable for Hackathon Presentation

Use modern software architecture standards and professional diagram notation.

Prompt 2: Verification based access to sensitive data

You are a Senior .NET Security Architect.

Update my existing .NET 8 MCP Server to support Sensitive Data Protection with Verification-Based Access Control.

OBJECTIVE

Normal company information should be accessible.

Sensitive information should require an additional verification step before access is granted.

DATA CLASSIFICATION

Public Data:
- Employee Name
- Department
- Project Name
- Work Email

Sensitive Data:
- Salary
- Personal Phone Number
- Home Address
- Performance Rating
- Bank Account Information

Restricted Data:
- Passwords
- Password Hashes
- API Keys
- Secrets
- Tokens

Restricted Data must NEVER be returned.

REQUIRED FLOW

User asks:

"Show Arun's salary"

MCP should:

1. Detect salary is sensitive
2. Check verification status
3. If not verified:
   Return:

   "Sensitive information requested.
   Verification required.
   Enter verification code."

4. User submits verification code

5. MCP validates code

6. If valid:
   Grant temporary access for 5 minutes

7. Return requested data

DATABASE CHANGES

Create table:

VerificationSessions

Columns:

SessionId
UserId
VerificationCode
ExpiresAt
IsVerified

NEW MCP TOOLS

Tool 1:

request_sensitive_access()

Purpose:
Generate verification code

Returns:
Verification code
Expiry time

Tool 2:

verify_sensitive_access(code)

Purpose:
Validate verification code

Returns:
Verified or Failed

Tool 3:

```
get_sensitive_employee_data(
    employeeName,
    dataType
)
```

Purpose:
Return sensitive information

Rules:

If verified:
Return data

If not verified:
Return verification required message

Tool 4:

get_employee_public_info(employeeName)

Purpose:
Return non-sensitive employee information

No verification required

SECURITY RULES

Never return:

- Password
- PasswordHash
- API Keys
- Tokens
- Secrets

Always return:

"Access denied. Restricted information."

ACCESS LEVELS

Employee
Manager
HR
Admin

Employee:
Can view public data only

Manager:
Can view team sensitive data after verification

HR:
Can view salary after verification

Admin:
Can view all sensitive data after verification

IMPLEMENTATION

Create:

AuthorizationService.cs
VerificationService.cs

Add:

Role validation
Verification validation
Session expiration
Audit logging

LOGGING

Log:

- Sensitive data requests
- Verification attempts
- Successful verifications
- Failed verifications
- Data access events

DEMO FLOW

User:
Show Arun salary

Claude:
Verification required

User:

Verify using code 123456

Claude:
Verification successful

User:
Show Arun salary

Claude:
Salary = ₹80,000

Generate complete code changes,
updated MCP tools,
database schema changes,
service classes,
and integration with existing MCP Server.